A database of exceptional graphs with least eigenvalue ≥ -2

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Abstract

There are exactly 54 732 619 exceptional graphs up to isomorphism, that is, connected graphs with least eigenvalue at least -2 that are not generalized line graphs. There are 54 732 046 with least eigenvalue -2, on 7 to 36 vertices, and 573 with least eigenvalue strictly greater than -2, on 6, 7 and 8 vertices. By theorems of Cameron, Goethals, Seidel and Shult, an exceptional graph is represented in the root system E_8 and has at most 36 vertices, so there are finitely many; Cvetković wrote in 2004 that their number is very big but not known. We prove that a connected graph with least eigenvalue at least -2 is represented in E_8 if and only if the rank of $2I + A$ is at most 8, with A the adjacency matrix, and that it is exceptional if and only if the pair formed by this rank and the determinant of the lattice spanned by its representation is $(6, 3)$, $(7, 2)$ or $(8, 1)$. We show how to find, by exact tests on integer matrices, every way of adding a vertex to a connected graph represented in E_8 resulting in a connected graph that is also represented in E_8. We then generate the 54 752 217 connected graphs represented in E_8, one vertex at a time, and thus produce a database of all exceptional graphs, up to isomorphism, which we make available. We obtained these results in the course of autoformalizing the paper of Cameron, Goethals, Seidel and Shult in the proof assistant Lean.

1 Introduction

We consider graphs with least eigenvalue at least -2, where by eigenvalues of a graph we mean those of its adjacency matrix. Line graphs have least eigenvalue at least -2, and so do the cocktail party graphs $CP(m)$ (K_{2m} minus a perfect matching). We recall that the *line graph* $L(Y)$ of a graph Y has the edges of Y as its vertices, two of them adjacent when they are incident with a common vertex of Y. For a graph Y on the vertices $1, \ldots, m$ and integers $a_1, \ldots, a_m \geq 0$, the *generalized line graph* $L(Y; a_1, \ldots, a_m)$, of [4, §4] and [8, §1.1], is obtained from $L(Y)$ by adding disjoint cocktail party graphs $CP(a_1), \ldots, CP(a_m)$, with every vertex of $CP(a_i)$ adjacent to every vertex of $L(Y)$ that, as an edge of Y, contains the vertex i. A graph X is *exceptional* if it is connected, its least eigenvalue $\lambda_{\min}(X)$ is at least -2, and X is not a generalized line graph [8, §1.1]. The classical result of Cameron, Goethals, Seidel and Shult [4] tells us that a connected graph with least eigenvalue at least -2 is either a generalized line graph or an exceptional graph, of which there are a finite number.

More specifically, they showed this by representing the exceptional graphs in the root system E_8. A *representation* of a graph X is a family of vectors $\mathbf{x}_u$ ($u \in V(X)$) in a real inner product

space with $\langle \mathbf{x}_u, \mathbf{x}_u \rangle = 2$ and $\langle \mathbf{x}_u, \mathbf{x}_w \rangle = A(u, w)$ for $u \neq w$, where $A = A(X)$ is the adjacency matrix of X. It is a representation in E_8 if every $\mathbf{x}_u$ is a root of E_8, and we say that X is *represented in* E_8 if it has such a representation; we will recall the root systems in Section 2. By [4, Theorem 4.10], a connected graph with least eigenvalue at least -2 is a generalized line graph or is represented in E_8; the theorem is stated for least eigenvalue -2 and leaves connectedness implicit, but its proof uses only that the graph is connected and that $\lambda_{\min} \geq -2$. Thus every exceptional graph is represented in E_8 and, by [4, Theorem 4.3], has at most 36 vertices. The only fact about generalized line graphs that our proofs use is [4, Theorem 4.2]: a graph is a generalized line graph if and only if it is represented in D_ℓ for some ℓ (Section 2).

The exceptional graphs with $\lambda_{\min} > -2$ are classical: there are 573 of them, 20 on 6 vertices, 110 on 7 and 443 on 8 [8, §2.3 and Table A2], [12]. The maximal exceptional graphs were found by Cvetković, Lepović, Rowlinson and Simić with the star complement technique: there are 473 of them, on 22 to 36 vertices [7], [8, Chapter 6]. In 2004, Cvetković wrote [6]:

> The set E of exceptional graphs is finite. The number of exceptional graphs is very big but not known. The set E can be partially ordered by the induced subgraph relation. In this order E has 20 minimal elements and 473 maximal ones.

In this note we determine the very big number.

Theorem 1.1. *There are exactly* 54 732 619 *exceptional graphs up to isomorphism:* 54 732 046 *with least eigenvalue* -2*, on* 7 *to* 36 *vertices, and* 573 *with least eigenvalue greater than* -2*, on* 6*,* 7 *and* 8 *vertices. For each* n*, the number of exceptional graphs on* n *vertices is given in Table 1.*

We also produce a database of all exceptional graphs (one for each isomorphism class), which we make available, with our code and the formalization in Lean, at `https://github.com/kguo-sagecode/least-eig-minus-2`.

The proof of Theorem 1.1 is a computation, but not a direct computation. The root system E_8 has 240 roots, which lie on 120 lines through the origin of $\mathbb{R}^8$ [4, Lemma 3.4], and the inner product of two roots on different lines is 0, 1 or -1. In a representation, distinct vertices have distinct roots, with inner product 0 or 1. Thus a representation in E_8 of a graph on n vertices is the same thing as a set of n lines of E_8 with a root chosen on each line so that no two chosen roots have inner product -1, and the graph is recovered from it: two vertices are adjacent when their roots have inner product 1. In principle, then, the exceptional graphs could be found by running through the sets of at most 36 of the 120 lines, with a root chosen on each line. But there are more than $4 \cdot 10^{41}$ such choices, and even the group of symmetries of E_8, which has order 696 729 600 [5, Chapter 4], reduces them to no fewer than $6 \cdot 10^{32}$ orbits. Since it is also computationally infeasible to go through the more than 10^{148} graphs on 36 vertices up to isomorphism, it was necessary to establish the lemmas of Sections 2 and 3, which make our enumeration feasible in four ways.

First, two integers decide whether a connected graph X with $\lambda_{\min}(X) \geq -2$ is represented in E_8 and whether it is exceptional. With $M(X) = 2I + A(X)$ and $\Lambda(X)$ the lattice spanned by a representation of X, the graph X is represented in E_8 if and only if $\operatorname{rk} M(X) \leq 8$, and it is exceptional if and only if the pair $(\operatorname{rk} M(X), \det \Lambda(X))$ is $(6, 3)$, $(7, 2)$ or $(8, 1)$ (Proposition 2.6). Both integers $(\operatorname{rk} M(X), \det \Lambda(X))$ are computed exactly. Note that we do not need to compute any eigenvalues, since $\lambda_{\min}(X) = -2$ if and only if $\operatorname{rk} M(X) < n$.

Second, a connected graph on at least two vertices represented in E_8 has a vertex whose deletion leaves a connected graph represented in E_8 (Lemma 3.1). Thus the connected graphs represented

in E_8 can be generated one vertex at a time, starting from K_1; we will call this process the *growth*, and the set $\mathcal{S}_n$ of the connected graphs on n vertices represented in E_8 a *level*. The levels contain generalized line graphs, since deleting a vertex of an exceptional graph can leave a generalized line graph; but they contain only those of rank at most 8, and this is what keeps their number small.

Third, the ways of adding a vertex to a graph X of a level, adjacent to at least one vertex of X, are found exactly. A certificate for X is a set B of $\operatorname{rk} M(X)$ vertices, so at most 8, that satisfies an identity of integer matrices (Definition 3.2), and the graphs obtained from X by adding such a vertex that are represented in E_8 correspond to the nonzero vectors $\mathbf{s} \in \{0,1\}^B$ that pass a test of integer identities and inequalities (Lemmas 3.4 and 3.5). Thus each graph has at most $2^8 - 1 = 255$ candidates, rather than the $2^n - 1$ nonempty sets of its vertices.

Fourth, isomorphic copies are removed by a canonical form, so that the growth with these tests gives exactly one graph from each isomorphism class of each level (Theorem 3.6). In all, the computation tests at most 255 candidates for each of 54 752 217 graphs, in exact integer arithmetic on matrices with at most 8 rows, and it takes less than an hour on 13 cores.

One could instead take the connected induced subgraphs of the 473 maximal exceptional graphs, since every exceptional graph is an induced subgraph of one of them; but the count would then rest on the completeness of the classification of [7]. The growth starts from K_1 and rests on no earlier classification; the maximal exceptional graphs come out of it (Section 4), and their known number is a check on it.

This note also serves as a companion to our formalization of [4] in the proof assistant Lean [11], and we obtained the enumeration in the course of it. The results of Sections 2 and 3 are proved in Lean as well; the computation is not, and the proof of Theorem 1.1 rests on the program that carries it out and on the checks that we made of the program and of its output. The appendices describe the program and why its tests are exact (Appendix A), the checks and the comparison with the literature (Appendix B), the database and how to regenerate it (Appendix C), and the formalization (Appendix D).

2 Root systems and the lattice criterion

All graphs are finite and simple. For a graph X on n vertices we will write $A = A(X)$ for its adjacency matrix, $\lambda_{\min}(X)$ for its least eigenvalue, and $M = M(X) = 2I + A$. Thus $\lambda_{\min}(X) \geq -2$ if and only if M is positive semidefinite, and $\lambda_{\min}(X) = -2$ if and only if, in addition, M is singular. We use functional notation, $M(u, w)$, for the (u, w) entry of a matrix M. For sets B and C of vertices, $M[B, C]$ is the submatrix of M with rows indexed by B and columns indexed by C; for convenience, we will write $\cdot$ for the whole vertex set and u for $\{u\}$. For a square matrix G, $\operatorname{adj}(G)$ is its adjugate, the transpose of its matrix of cofactors, which satisfies $G \operatorname{adj}(G) = \operatorname{adj}(G)\, G = (\det G) I$ and has integer entries when G does. We will write $\mathbf{e}_1, \ldots, \mathbf{e}_\ell$ for the standard basis of $\mathbb{R}^\ell$. For a statement P, we will write $[P]$ for 1 if P holds and 0 otherwise, and the *norm* of a vector $\mathbf{x}$ is $\langle \mathbf{x}, \mathbf{x} \rangle$.

In this section we will prove the lattice criterion of Section 1, Proposition 2.6. We will use the terminology of [4]. A set of lines through the origin of a real inner product space, any two at 60° or 90°, is *star-closed* if with any two of its lines at 60° it contains the third line of their star, the line in their plane at 60° to both; it is *indecomposable* if it is not the union of two sets of lines lying in orthogonal proper subspaces of its span. By [4, Theorem 3.5], the indecomposable star-closed sets of lines at 60° and 90° are, up to isometry, the *root systems* A_r ($r \geq 1$), D_r ($r \geq 4$), E_6, E_7 and E_8,

which [4, §3] gives as explicit sets of lines; the *rank* of a root system is the dimension of its span, and its *roots* are the vectors of norm 2 on its lines, two on each line. We will write D_ℓ also for the model of [4, §3], the lines spanned by the vectors $\pm\mathbf{e}_i \pm \mathbf{e}_j$ $(i \neq j)$ of $\mathbb{R}^\ell$, which is defined for every $\ell \geq 2$ and is the root system D_ℓ for $\ell \geq 4$; [4, Theorem 4.2] is stated for these models. Extending the definition of Section 1, we will say that a representation $(\mathbf{x}_u)$ of a graph X is a representation *in* a root system R if every $\mathbf{x}_u$ is a root of R, and that X is *represented in* R if it has such a representation. The Gram matrix of a representation of X is $M(X)$, and X has a representation if and only if $M(X)$ is positive semidefinite, since a real symmetric matrix is positive semidefinite if and only if it is a Gram matrix. If $(\mathbf{x}_u)$ and $(\mathbf{x}'_u)$ are representations of X, they have the same Gram matrix, so the map $\mathbf{x}_u \mapsto \mathbf{x}'_u$ extends to a linear isometry of their spans. We will write $\Lambda(X)$ for the $\mathbb{Z}$-span of a representation; up to isometry it depends only on X. Inner products of its vectors are integers, since those of the $\mathbf{x}_u$ are, so a nonzero vector of $\Lambda(X)$ has positive integer norm. Thus $\Lambda(X)$ is discrete, and it is an integral lattice whose rank is the dimension $r = \operatorname{rk} M(X)$ of the span of the representation. We will write $\det \Lambda(X)$ for the determinant of the Gram matrix of a $\mathbb{Z}$-basis of $\Lambda(X)$; it does not depend on the basis, since a change of basis has determinant ± 1.

The first lemma is the observation of [4, §4] that a representation of a connected graph lies in an indecomposable star-closed set of lines, in the form in which we will need it.

Lemma 2.1. *Let X be a connected graph with $\lambda_{\min}(X) \geq -2$, let $(\mathbf{x}_u)$ be a representation of X, and let Φ be the set of vectors of norm 2 in $\Lambda(X)$. Then the lines spanned by the vectors of Φ form an indecomposable star-closed set of lines at $60°$ and $90°$ whose span is the span of the representation, Φ is the set of roots of this set of lines, and every $\mathbf{x}_u$ lies in Φ.*

Proof. Inner products of vectors of $\Lambda(X)$ are integers, and $\Lambda(X) = -\Lambda(X)$, so $\Phi = -\Phi$. Let $\mathbf{y}$ and $\mathbf{z}$ be vectors of Φ on distinct lines. By the Cauchy–Schwarz inequality $|\langle \mathbf{y}, \mathbf{z}\rangle| < 2$, since equality would make $\mathbf{y}$ and $\mathbf{z}$ linearly dependent, so $\langle \mathbf{y}, \mathbf{z}\rangle$ is 0, 1 or -1, and the two lines are at $90°$ or $60°$. The set of lines spanned by the vectors of Φ is finite, since $\Lambda(X)$ is discrete. It is star-closed: if two of its lines are at $60°$, we can choose $\mathbf{y}$ and $\mathbf{z}$ in Φ on them with $\langle \mathbf{y}, \mathbf{z}\rangle = -1$, and then $\mathbf{y} + \mathbf{z}$ lies in $\Lambda(X)$ and has norm $2 - 2 + 2 = 2$, so it lies in Φ, and its line is the third line of their star, since $\langle \mathbf{y} + \mathbf{z}, \mathbf{y}\rangle = \langle \mathbf{y} + \mathbf{z}, \mathbf{z}\rangle = 1$. The vectors of norm 2 on the line of $\mathbf{y}$ are $\pm\mathbf{y}$, both in Φ, so Φ is the set of roots of this set of lines. Each $\mathbf{x}_u$ has norm 2 and lies in $\Lambda(X)$, so it lies in Φ, and $\Phi \subseteq \Lambda(X)$ lies in the span of the representation; thus the span of Φ is the span of the representation. Finally, suppose that the lines of Φ are the union of two sets of lines lying in orthogonal proper subspaces U and W of this span. Each $\mathbf{x}_u$ lies in U or in W, and adjacent vertices u and w have $\langle \mathbf{x}_u, \mathbf{x}_w\rangle = 1$, so $\mathbf{x}_u$ and $\mathbf{x}_w$ lie in the same one of the two. Since X is connected, all the $\mathbf{x}_u$ lie in one of them, say U, and then U contains their span and is not proper. Thus the set of lines is indecomposable. □

We will use the following facts from the theory of root systems [14, Chapter III]. A root system of rank r has a set of r *simple roots*, a basis of its span, any two distinct ones of which have inner product 0 or -1; its *Dynkin diagram* has the simple roots as nodes, two of them adjacent when their inner product is -1, and its *Cartan matrix* is the Gram matrix of the simple roots. The *root lattice* of a root system is the $\mathbb{Z}$-span of its roots; the simple roots are a $\mathbb{Z}$-basis of it, so its determinant is the determinant of the Cartan matrix.

Lemma 2.2. *Let X be a connected graph with $\lambda_{\min}(X) \geq -2$, and let $r = \operatorname{rk} M(X)$. Then the set Φ of vectors of norm 2 in $\Lambda(X)$ is the set of roots of an indecomposable root system of rank r, of type A_r $(r \geq 1)$, D_r $(r \geq 4)$, E_6, E_7 or E_8, and $\Lambda(X)$ is its root lattice. In particular $\det \Lambda(X)$ is $r + 1$, 4, 3, 2 or 1, respectively.*

Proof. By Lemma 2.1, Φ is the set of roots of an indecomposable star-closed set of lines whose span is the span of the representation, which has dimension r. By [4, Theorem 3.5] it is one of the listed root systems, of rank r. The vectors $\mathbf{x}_u$ lie in Φ and generate $\Lambda(X)$, and $\Phi \subseteq \Lambda(X)$, so $\Lambda(X)$ is the $\mathbb{Z}$-span of Φ, which is the root lattice. The determinants are the determinants of the Cartan matrices [14, §13.1], [5, Chapter 4]. □

We will compute $\det \Lambda(X)$ in two independent ways: the growth uses Lemma 2.4, and the verification of Appendix B uses Lemma 2.3. The *elementary divisors* of an integer matrix are the diagonal entries of its Smith normal form.

Lemma 2.3. *Let X be a graph with $\lambda_{\min}(X) \geq -2$ and $r = \operatorname{rk} M(X)$. Then $\det \Lambda(X)$ is the product of the r nonzero elementary divisors of $M(X)$.*

Proof. Let $\mathbf{v}_1, \dots, \mathbf{v}_r$ be a basis of $\Lambda = \Lambda(X)$, with Gram matrix F, so that $\det F = \det \Lambda$, and let U be the $r \times n$ integer matrix with $\mathbf{x}_u = \sum_i U(i,u)\mathbf{v}_i$. Then $M = U^T F U$. The columns of U generate $\mathbb{Z}^r$, since the $\mathbf{x}_u$ generate Λ, so the elementary divisors of U are all 1, and $U = P\,[I_r\ 0]\,Q$ with $P \in GL_r(\mathbb{Z})$ and $Q \in GL_n(\mathbb{Z})$. Thus

$$M = Q^T \begin{pmatrix} P^T F P & 0 \\ 0 & 0 \end{pmatrix} Q,$$

and the elementary divisors of M are those of the nonsingular matrix $P^T F P$ together with $n - r$ zeros. The product of the elementary divisors of $P^T F P$ is $|\det(P^T F P)| = \det F$, and the result follows. □

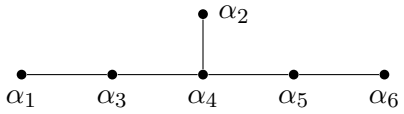
Lemma 2.4. *Let X be a graph with $\lambda_{\min}(X) \geq -2$, and let B be a set of $r = \operatorname{rk} M(X)$ vertices with $G = M[B,B]$ nonsingular; write $D = \det G$ and $C = \operatorname{adj}(G)\, M[B,\cdot]$, an integer $r \times n$ matrix. Let H be the subgroup of $(\mathbb{Z}/D\mathbb{Z})^r$ generated by the columns of C taken modulo D. Then $|H|^2$ divides D and $\det \Lambda(X) = D/|H|^2$.*

Proof. The $\mathbf{x}_b$ ($b \in B$) are a basis of the span, since their Gram matrix G is nonsingular. In these coordinates $\mathbf{x}_u$ has coordinate vector $G^{-1}M[B,u] = C(\cdot,u)/D$, since $M[B,u]$ is the vector of inner products of $\mathbf{x}_u$ with the $\mathbf{x}_b$. Thus the sublattice Λ_B spanned by the $\mathbf{x}_b$ is $\mathbb{Z}^r$ in these coordinates, and $\Lambda(X) = \mathbb{Z}^r + \sum_u \mathbb{Z}\, C(\cdot,u)/D$. The map $\mathbf{y} \mapsto D\mathbf{y} \bmod D$ from $\Lambda(X)$ to $(\mathbb{Z}/D\mathbb{Z})^r$ has kernel $\mathbb{Z}^r = \Lambda_B$ and image H, so $\Lambda(X)/\Lambda_B \cong H$. We obtain that $D = \det \Lambda_B = [\Lambda(X) : \Lambda_B]^2 \det \Lambda(X) = |H|^2 \det \Lambda(X)$, where $\det \Lambda(X)$ is an integer since $\Lambda(X)$ is integral. □

We will use the next lemma in the proof of Proposition 2.6, to show that the lattice of a generalized line graph is not of type E. It is due to Mordell [17], who showed that the quadratic form of E_6 is not a sum of squares of linear forms with integer coefficients; a linear isometry from the root lattice of E_6 into $\mathbb{Z}^m$ would write this form as a sum of m such squares. We will include a proof for completeness.

Lemma 2.5 ([17])**.** *There is no linear isometry from the root lattice of E_6 into $\mathbb{Z}^m$, for any m.*

Proof. Suppose that ϕ is one. The vectors of norm 2 of $\mathbb{Z}^m$ are the $\pm\mathbf{e}_i \pm \mathbf{e}_j$ ($i \neq j$), so ϕ maps the simple roots $\alpha_1, \dots, \alpha_6$ of E_6 to vectors of this form. We use the labelling of Bourbaki: α_4 is the branch node (the node of valency 3), its neighbours are $\alpha_2, \alpha_3, \alpha_5$, and $\alpha_1 \sim \alpha_3$, $\alpha_6 \sim \alpha_5$, where adjacent simple roots have inner product -1 and the others 0; the Dynkin diagram is



After a signed permutation of the coordinates we may assume that $\phi(\alpha_4) = \mathbf{e}_1 - \mathbf{e}_2$. Two vectors of the form $\pm\mathbf{e}_i \pm \mathbf{e}_j$ have inner product ± 1 if and only if they share exactly one coordinate. Thus each of $\phi(\alpha_2), \phi(\alpha_3), \phi(\alpha_5)$ is $-\mathbf{e}_1 \pm \mathbf{e}_k$ or $\mathbf{e}_2 \pm \mathbf{e}_k$ with $k \notin \{1, 2\}$. Two of them, say $\mathbf{y}$ and $\mathbf{z}$, involve the same one of the coordinates 1 and 2, say the coordinate 1 (the other case is the same after the isometry $\mathbf{e}_1 \leftrightarrow -\mathbf{e}_2$). Then $\mathbf{y} = -\mathbf{e}_1 + \epsilon\mathbf{e}_k$ and $\mathbf{z} = -\mathbf{e}_1 + \delta\mathbf{e}_l$ with $0 = \langle \mathbf{y}, \mathbf{z}\rangle = 1 + \epsilon\delta[k = l]$, so $l = k$ and $\mathbf{z} = -\mathbf{e}_1 - \epsilon\mathbf{e}_k$. Of $\alpha_2, \alpha_3, \alpha_5$, only α_2 is an end node (a node of valency 1), so one of $\mathbf{y}$, $\mathbf{z}$, say $\mathbf{y} = \phi(\beta)$, is the image of α_3 or α_5; let γ be α_1 or α_6 respectively, so that γ is adjacent to β and orthogonal to α_4 and to the preimage of $\mathbf{z}$. Now $\phi(\gamma)$ shares exactly one coordinate with $\mathbf{y}$. If it involves the coordinate 1, then $\langle \phi(\gamma), \mathbf{e}_1 - \mathbf{e}_2\rangle = 0$ forces $\phi(\gamma) = \pm(\mathbf{e}_1 + \mathbf{e}_2)$, and then $\langle \phi(\gamma), \mathbf{z}\rangle = \mp 1 \neq 0$. Otherwise $\phi(\gamma) = -\epsilon\mathbf{e}_k \pm \mathbf{e}_j$ with $j \notin \{1, k\}$, and then $\langle \phi(\gamma), \mathbf{z}\rangle = \epsilon^2 = 1 \neq 0$. Both contradict the orthogonality of γ and the preimage of $\mathbf{z}$. □

Proposition 2.6. *Let X be a connected graph with $\lambda_{\min}(X) \geq -2$, let $r = \operatorname{rk} M(X)$, and let Φ be as in Lemma 2.2.*

(1) X is a generalized line graph if and only if Φ has type A_r or D_r.

(2) X is exceptional if and only if $(r, \det \Lambda(X))$ is one of $(6, 3)$, $(7, 2)$ and $(8, 1)$, that is, if and only if Φ has type E_6, E_7 or E_8.

(3) X is represented in E_8 if and only if $r \leq 8$.

Proof. For (1), suppose first that Φ has type A_r or D_r. We will write $\Phi(R)$ for the set of roots of the model of a root system R in [4, §3]. The isometry of [4, Theorem 3.5] maps Φ onto $\Phi(\mathrm{A}_r)$ or $\Phi(\mathrm{D}_r)$, and $\Phi(\mathrm{A}_r) \subseteq \Phi(\mathrm{D}_{r+1})$, so X is represented in D_ℓ for some ℓ, and it is a generalized line graph by [4, Theorem 4.2]; this is the argument of [4, Theorem 4.10]. Conversely, let $X \cong L(Y; a_1, \dots, a_m)$. By [4, Theorem 4.2], X has a representation $(\mathbf{y}_u)$ in $\mathrm{D}_\ell \subseteq \mathbb{Z}^\ell$. It has Gram matrix $M(X)$, so $\Lambda(X)$ is isometric to the $\mathbb{Z}$-span of the $\mathbf{y}_u$, which lies in $\mathbb{Z}^\ell$. If Φ had type E_k, then $\Lambda(X)$ would contain the root lattice of E_6 (the Dynkin diagram of E_6 is a subdiagram of those of E_7 and E_8), contradicting Lemma 2.5. Thus Φ has type A_r or D_r.

Part (2) follows from (1) and Lemma 2.2: E_r exists only for $r \in \{6, 7, 8\}$, with determinant $9 - r$, and for these r the determinants $r + 1$, 4 and $9 - r$ of A_r, D_r and E_r are distinct (for $r = 6$, 7 and 8 they are $7, 4, 3$; $8, 4, 2$; and $9, 4, 1$).

For (3), a representation in E_8 spans a space of dimension at most 8, so $r \leq 8$. Conversely, let $r \leq 8$. Each of A_r, D_r ($r \leq 8$), E_6, E_7, E_8 is isometric to a subset of $\Phi(\mathrm{E}_8)$. Indeed, take the model of E_8 in $\mathbb{R}^8$ whose roots are the vectors $\pm\mathbf{e}_i \pm \mathbf{e}_j$ ($i \neq j$) and the vectors $\frac{1}{2}(\pm 1, \dots, \pm 1)$ with an even number of minus signs. The roots $\pm\mathbf{e}_i \pm \mathbf{e}_j$ form D_8, which contains the model D_r for $r \leq 8$, and $\mathrm{A}_r \subseteq \mathrm{D}_{r+1} \subseteq \mathrm{D}_8$ for $r \leq 7$. For A_8, the vectors $\frac{1}{2}(-1, 1, 1, 1, 1, 1, 1, 1)$ and $\mathbf{e}_i - \mathbf{e}_{i+1}$ ($1 \leq i \leq 7$) are simple roots of an A_8 in the model with an odd number of minus signs, and negating a coordinate maps that model onto the first one. Finally $\mathrm{E}_6 \subseteq \mathrm{E}_7 \subseteq \mathrm{E}_8$ [4, §3]. Composing with the isometry of [4, Theorem 3.5], X is represented in E_8. □

3 Growing the graphs represented in E_8

Recall that, for $n \geq 1$, the level $\mathcal{S}_n$ is the set of isomorphism classes of connected graphs on n vertices represented in E_8, that is, by Proposition 2.6, with M positive semidefinite of rank at most 8; we will say that a graph is in $\mathcal{S}_n$ when its isomorphism class is. By [4, Theorem 4.3], $\mathcal{S}_n = \varnothing$ for $n \geq 37$.

Lemma 3.1. *Let $n \geq 1$, and let Z be a connected graph on $n+1$ vertices. Then Z has a vertex v such that $Z - v$, the graph obtained by deleting v and its edges, is connected, and v has a neighbour. If Z is represented in E_8, then so is $Z - v$. Thus every graph in $\mathcal{S}_{n+1}$ arises from a graph in $\mathcal{S}_n$ by adding a vertex adjacent to a nonempty set of vertices.*

Proof. A leaf of a spanning tree is such a vertex, and it has a neighbour since Z is connected and has at least 2 vertices. A representation of Z in E_8 restricts to one of $Z - v$. □

Definition 3.2. Let X be a graph on n vertices and $M = M(X)$. A *certificate* for X is a set B of vertices such that, in some ordering of B, the matrix $G = M[B, B]$ has all its leading principal minors positive, and

$$D\, M = M[\cdot, B]\, \mathrm{adj}(G)\, M[B, \cdot], \qquad D = \det G.$$

By Sylvester's criterion, the first condition says that G is positive definite, so it does not depend on the ordering. Both conditions are statements about integer matrices, which we can check exactly.

Lemma 3.3. *Let X be a graph and $M = M(X)$. Then X has a certificate if and only if M is positive semidefinite, and every certificate for X has exactly $\mathrm{rk}\, M$ vertices. Further, if M is positive semidefinite, then any $\mathrm{rk}\, M$ vertices whose rows of M are independent form a certificate.*

Proof. If B is a certificate, then G is positive definite and $M = M[\cdot, B]\, G^{-1} M[B, \cdot]$, so M is positive semidefinite of rank $|B|$, since $M[B, \cdot]$ has rank $|B|$. Conversely, let M be positive semidefinite of rank r, and let B be a set of r vertices whose rows of M are independent. Then M is the Gram matrix of vectors $\mathbf{x}_u$, and the $\mathbf{x}_b$ ($b \in B$) are independent since their rows of M are, so G is positive definite and its leading principal minors are positive. Every row $M[u, \cdot]$ is a combination $\mathbf{w}^T M[B, \cdot]$ of the rows in B, and restricting to the columns in B gives $\mathbf{w}^T = M[u, B]\, G^{-1}$. Thus $M = M[\cdot, B]\, G^{-1} M[B, \cdot]$, which is the identity of Definition 3.2, since $\mathrm{adj}(G) = D\, G^{-1}$. □

For a graph X and a set N of vertices of X, we will write X_N for the graph obtained from X by adding a vertex v adjacent to exactly the vertices in N.

Lemma 3.4. *Let X be a connected graph on n vertices with a certificate B of size $r \leq 8$, and let G, D be as in Definition 3.2 and $C = \mathrm{adj}(G)\, M[B, \cdot]$. For a nonempty set N of vertices of X, let $\mathbf{s} \in \{0, 1\}^B$ be the indicator vector of $N \cap B$. Then X_N is represented in E_8 if and only if*

(1) $\mathbf{s}^T C(\cdot, u) = D\, [u \in N]$ for every vertex u of X, and

(2) $\mathbf{s}^T\, \mathrm{adj}(G)\, \mathbf{s} \leq 2D$, with equality if $r = 8$.

Proof. X_N is connected since X is and $N \neq \varnothing$. Let $(\mathbf{x}_u)$ be a representation of X in $\mathbb{R}^r$, so that the $\mathbf{x}_b$ ($b \in B$) are a basis. Suppose first that X_N is represented in E_8. Then $M(X_N)$ is the Gram matrix of vectors of $\mathbb{R}^8$, so X_N has a representation, and $M(X_N)$ has rank at most 8. The vectors of this representation are those of a representation of X, which span a space of dimension r, and

one more vector, so they span a space of dimension at most $r+1$. Thus X_N has a representation in $\mathbb{R}^{r+1} \supseteq \mathbb{R}^r$, and after an isometry of $\mathbb{R}^{r+1}$ we may assume that it extends $(\mathbf{x}_u)$, with the new vector $\mathbf{x}_v$. Let $\mathbf{p}$ be the orthogonal projection of $\mathbf{x}_v$ onto $\mathbb{R}^r$. Then $\langle \mathbf{p}, \mathbf{x}_b \rangle = \mathbf{s}(b)$, so $\mathbf{p} = \sum_b (G^{-1}\mathbf{s})(b)\mathbf{x}_b$, and, since $\mathbf{x}_v - \mathbf{p}$ is orthogonal to every $\mathbf{x}_u$, $\langle \mathbf{x}_v, \mathbf{x}_u \rangle = \langle \mathbf{p}, \mathbf{x}_u \rangle = \mathbf{s}^T G^{-1} M[B,u] = \mathbf{s}^T C(\cdot, u)/D$; this is condition 1, and it holds whether or not $\mathbf{x}_v$ lies in $\mathbb{R}^r$. Thus the neighbours of v in X are determined by $\mathbf{s}$. Further $\langle \mathbf{p}, \mathbf{p} \rangle = \mathbf{s}^T G^{-1} \mathbf{s} \leq \langle \mathbf{x}_v, \mathbf{x}_v \rangle = 2$, with equality if and only if $\mathbf{x}_v \in \mathbb{R}^r$, which is forced when $r = 8$ since the rank is at most 8; this is condition 2, since $\operatorname{adj}(G) = D\,G^{-1}$. Conversely, if both conditions hold, let $\mathbf{p} = \sum_b (G^{-1}\mathbf{s})(b)\mathbf{x}_b$, let $\mathbf{z}$ be a unit vector orthogonal to $\mathbb{R}^r$ in $\mathbb{R}^{r+1}$, and let $\mathbf{x}_v = \mathbf{p} + \sqrt{2 - \langle \mathbf{p}, \mathbf{p} \rangle}\,\mathbf{z}$. Then $\langle \mathbf{x}_v, \mathbf{x}_v \rangle = 2$ and, by condition 1, $\langle \mathbf{x}_v, \mathbf{x}_u \rangle = [u \in N]$, so we have a representation of X_N whose span has dimension r or $r+1$, and at most 8 by condition 2. By Proposition 2.6, X_N is represented in E_8. □

Condition 1 involves N, while the growth will run over the vectors $\mathbf{s}$; we will now recover N from $\mathbf{s}$. With X, B, G, D and C as in Lemma 3.4, we will call a vector $\mathbf{s} \in \{0,1\}^B$ *admissible* if $\mathbf{s}^T C(\cdot, u) \in \{0, D\}$ for every vertex u of X and condition 2 holds, and we will write $N_\mathbf{s}$ for the set of vertices u with $\mathbf{s}^T C(\cdot, u) = D$.

Lemma 3.5. *Let X be a connected graph with a certificate B of size $r \leq 8$, and let G, D and C be as in Lemma 3.4. Then the nonempty sets N of vertices of X with X_N represented in E_8 are exactly the sets $N_\mathbf{s}$ for the nonzero admissible vectors $\mathbf{s} \in \{0,1\}^B$. Distinct vectors give distinct sets, and the vector $\mathbf{s}$ with $N_\mathbf{s} = N$ is the indicator vector of $N \cap B$.*

Proof. Since $\operatorname{adj}(G)\,G = DI$, we have $C(b', b) = D\,[b' = b]$ for $b, b' \in B$, so $\mathbf{s}^T C(\cdot, b) = D\,\mathbf{s}(b)$, and $\mathbf{s}$ is the indicator vector of $N_\mathbf{s} \cap B$; in particular, distinct vectors give distinct sets. If $\mathbf{s}$ is admissible, then conditions 1 and 2 of Lemma 3.4 hold for $N_\mathbf{s}$ and $\mathbf{s}$, and $N_\mathbf{s} \neq \varnothing$ when $\mathbf{s} \neq 0$; thus $X_{N_\mathbf{s}}$ is represented in E_8 when $\mathbf{s}$ is nonzero and admissible. Conversely, if X_N is represented in E_8 for a nonempty set N, then conditions 1 and 2 hold for N and the indicator vector $\mathbf{s}$ of $N \cap B$, by Lemma 3.4. Then $\mathbf{s}$ is admissible and $N = N_\mathbf{s}$, and $\mathbf{s} \neq 0$, since otherwise condition 1 gives $N = \varnothing$. □

We will call the graphs $X_{N_\mathbf{s}}$, for the nonzero admissible vectors $\mathbf{s}$, the *children* of X. By Lemma 3.5, they are the graphs X_N represented in E_8, for the nonempty sets N of vertices of X, so they do not depend on the choice of the certificate B.

A *canonical form* assigns to each graph a graph isomorphic to it, in such a way that two graphs have the same canonical form if and only if they are isomorphic; the computation uses the canonical form of a canonical labelling program (see Appendix A). The next theorem shows that the growth misses no graph and produces only graphs of the levels.

Theorem 3.6. *Let $\mathcal{L}_1 = \{K_1\}$ and, for $n \geq 1$, let $\mathcal{L}_{n+1}$ be the set of canonical forms of the children of the graphs $X \in \mathcal{L}_n$. Then every connected graph on n vertices represented in E_8 is isomorphic to a graph in $\mathcal{L}_n$, and every graph in $\mathcal{L}_n$ is connected and represented in E_8.*

Proof. By induction on n; the case $n = 1$ is clear, since $M(K_1) = (2)$. Each graph in $\mathcal{L}_{n+1}$ is connected and represented in E_8, since the children of a connected graph are connected and, by Lemma 3.5, represented in E_8. Conversely let Z be a connected graph on $n+1$ vertices represented in E_8. By Lemma 3.1 it has a vertex v with $Z - v$ connected and represented in E_8, so by induction there are $X \in \mathcal{L}_n$ and an isomorphism $\phi\colon Z - v \to X$. Let $N = \phi(N_Z(v))$, where $N_Z(v)$ is the set of neighbours of v in Z; it is nonempty since Z is connected. Then $X_N \cong Z$ is represented in E_8, so X_N is a child of X by Lemma 3.5, and its canonical form is a graph of $\mathcal{L}_{n+1}$ isomorphic to Z. □

Since canonical forms are equal exactly for isomorphic graphs, two graphs of $\mathcal{L}_n$ are isomorphic only if they are equal; thus the growth does not produce the same graph twice, and $\mathcal{L}_n$ consists of exactly one graph from each class of $\mathcal{S}_n$. We can now prove Theorem 1.1.

Proof of Theorem 1.1. Let $n \geq 1$. By Proposition 2.6, the exceptional graphs on n vertices are the graphs in $\mathcal{S}_n$ with $(\operatorname{rk} M, \det \Lambda) \in \{(6,3),(7,2),(8,1)\}$; those with $\lambda_{\min} = -2$ are those with $\operatorname{rk} M < n$, and those with $\lambda_{\min} > -2$ are those with $\operatorname{rk} M = n$. Since $\mathcal{L}_n$ consists of exactly one graph from each class of $\mathcal{S}_n$, the number of exceptional graphs on n vertices with $\lambda_{\min} = -2$, up to isomorphism, is the number of graphs $X \in \mathcal{L}_n$ with $(\operatorname{rk} M(X), \det \Lambda(X))$ one of the three pairs and $\operatorname{rk} M(X) < n$, and the number with $\lambda_{\min} > -2$ is the number of such graphs with $\operatorname{rk} M(X) = n$. The program of Appendix A computes the sets $\mathcal{L}_1, \ldots, \mathcal{L}_{37}$: for each graph X of each $\mathcal{L}_n$, it finds a certificate B, so that $\operatorname{rk} M(X) = |B|$ by Lemma 3.3, then the nonzero admissible vectors, whose graphs $X_{N_\mathbf{s}}$ are the children of X, and, when $|B| \geq 6$, $\det \Lambda(X)$ by Lemma 2.4, all in exact integer arithmetic. It finds $\mathcal{L}_{37} = \varnothing$, as it must, since $\mathcal{S}_{37} = \varnothing$. It writes the exceptional graphs of each $\mathcal{L}_n$ to the files of the database, those with $\operatorname{rk} M(X) < n$ and those with $\operatorname{rk} M(X) = n$ to separate files (Theorem C.1), and it records the number of graphs in each file; Table 1 is made from this record, which the verification of Appendix B checks against the files. An exceptional graph has $6 \leq \operatorname{rk} M \leq 8$ and $\operatorname{rk} M \leq n$, so there is none on fewer than 6 vertices, $\operatorname{rk} M = n$ only for $n \in \{6,7,8\}$, and $\operatorname{rk} M < n$ only for $n \geq 7$; and there is none on more than 36 vertices, since $\mathcal{S}_n = \varnothing$ for $n \geq 37$. □

An exceptional graph X is *maximal* if it is not an induced subgraph of a larger exceptional graph. An exceptional graph is connected, and by Lemma 3.3 and Proposition 2.6 its certificates have $\operatorname{rk} M \leq 8$ vertices, so admissible vectors are defined for it.

Proposition 3.7. *Let X be an exceptional graph, and let B be a certificate for X. Then X is maximal if and only if no nonzero vector of $\{0,1\}^B$ is admissible, that is, if and only if no graph X_N, for a nonempty set N of vertices of X, is represented in* E_8.

Proof. The two conditions are equivalent by Lemma 3.5. First, an induced subgraph of a generalized line graph is a generalized line graph: by [4, Theorem 4.2] a graph is a generalized line graph if and only if it is represented in D_ℓ for some ℓ, and a representation in D_ℓ restricts to one of every induced subgraph. Suppose that X is not maximal, and let Z be an exceptional graph containing X properly as an induced subgraph. Since Z is connected, it has a vertex v outside X with a neighbour in X, and the subgraph of Z induced on $V(X) \cup \{v\}$ is X_N for the nonempty set N of neighbours of v in X. Its matrix M is a principal submatrix of $M(Z)$, so it is positive semidefinite; X_N is connected, and it is not a generalized line graph, since its induced subgraph X is not. Thus X_N is exceptional, so it is represented in E_8 by Proposition 2.6. Conversely, if X_N is represented in E_8 for a nonempty set N, then X_N is connected and $\lambda_{\min}(X_N) \geq -2$, and it is not a generalized line graph, since its induced subgraph X is not. Thus X_N is an exceptional graph containing X properly, and X is not maximal. □

4 The counts by number of vertices

Table 1 gives the counts for each number n of vertices. As there and in the database (Appendix C), we will write *list (A)* for the exceptional graphs with $\lambda_{\min} = -2$ and *list (B)* for those with $\lambda_{\min} > -2$.

n	$\lvert\mathcal{S}_n\rvert$	(A) $\lambda_{\min} = -2$	(B) $\lambda_{\min} > -2$	exceptional	maximal
1	1	0	0	0	0
2	1	0	0	0	0
3	2	0	0	0	0
4	6	0	0	0	0
5	18	0	0	0	0
6	67	0	20	20	0
7	291	51	110	161	0
8	1 410	585	443	1 028	0
9	6 428	5 594	0	5 594	0
10	25 584	24 097	0	24 097	0
11	87 484	85 341	0	85 341	0
12	257 048	254 444	0	254 444	0
13	649 914	647 196	0	647 196	0
14	1 418 205	1 415 651	0	1 415 651	0
15	2 677 814	2 675 647	0	2 675 647	0
16	4 385 365	4 383 688	0	4 383 688	0
17	6 238 430	6 237 257	0	6 237 257	0
18	7 714 028	7 713 273	0	7 713 273	0
19	8 288 067	8 287 627	0	8 287 627	0
20	7 724 697	7 724 461	0	7 724 461	0
21	6 227 347	6 227 227	0	6 227 227	0
22	4 324 420	4 324 362	0	4 324 362	1
23	2 572 833	2 572 808	0	2 572 808	0
24	1 303 562	1 303 551	0	1 303 551	0
25	559 586	559 581	0	559 581	0
26	203 365	203 363	0	203 363	0
27	63 212	63 211	0	63 211	0
28	17 245	17 244	0	17 244	1
29	4 367	4 367	0	4 367	432
30	1 041	1 041	0	1 041	25
31	272	272	0	272	7
32	75	75	0	75	3
33	22	22	0	22	1
34	7	7	0	7	2
35	2	2	0	2	0
36	1	1	0	1	1
total	54 752 217	54 732 046	573	54 732 619	473

Table 1: The numbers of graphs on n vertices that are connected and represented in E_8 (the size of the level $\mathcal{S}_n$), exceptional with $\lambda_{\min} = -2$ (list (A)), exceptional with $\lambda_{\min} > -2$ (list (B)), exceptional, and maximal exceptional.

As we saw after Theorem 3.6, $|\mathcal{S}_n| = |\mathcal{L}_n|$, and the program of Appendix A records the number of graphs of each $\mathcal{L}_n$ as it writes the level. Thus there are exactly 54 752 217 connected graphs represented in E_8 up to isomorphism, none of them on more than 36 vertices [4, Theorem 4.3]. By Proposition 3.7, an exceptional graph with a certificate B is maximal if and only if no nonzero vector of $\{0,1\}^B$ is admissible; the program records the number of nonzero admissible vectors of every graph of the levels, and the exceptional graphs for which this number is 0, the maximal exceptional graphs, are 473 graphs on 22 to 36 vertices. These numbers agree with those of [7] and [6], and the numbers of list (B), of the *minimal* exceptional graphs (those with no proper induced subgraph that is exceptional) and of the regular exceptional graphs agree with the literature as well; Appendix B gives the comparisons.

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A The program and the exactness of its tests

Exactness

The program `grow.py`, in the directory `exceptional/database/scripts/` of the repository, carries out the construction of Theorem 3.6 and the classification of Proposition 2.6. Floating point is used only to propose B and $\mathrm{adj}(G)$, and each proposal is then checked in exact arithmetic.

For each graph the program proposes a set B with the pivoted Cholesky factorization of LAPACK [1] (`dpstrf` through SciPy [21]), and then checks that B is a certificate. First, it checks that the leading principal minors of G are positive, by fraction-free (Bareiss) elimination in the integers of Python [19], which gives $D = \det G$ exactly, as the last of them. Next, it proposes $\mathrm{adj}(G)$: the matrix T obtained by rounding $D\,G^{-1}$, computed in floating point, is accepted if its entries are finite and at most 2^{r-1} in absolute value and $GT = DI$ holds in 64-bit integers, and otherwise $\mathrm{adj}(G)$ is computed exactly, by Gauss-Jordan elimination over the rationals. The bound is the one that $\mathrm{adj}(G)$ satisfies: G is positive definite with diagonal entries 2, so $\mathrm{adj}(G) = D\,G^{-1}$ is positive definite, its diagonal entries are principal minors of G, at most 2^{r-1} by Hadamard's inequality, and its other entries are bounded by its diagonal entries. Under the bound the product GT cannot overflow, and $GT = DI$ gives $T = \mathrm{adj}(G)$, since G is nonsingular. Finally, it checks the identity of Definition 3.2 in 64-bit integers; with $r \le 8$ for the graphs of the levels, the entries of $\mathrm{adj}(G)$ are at most 2^7 and those of the two sides of the identity at most 2^{15} in absolute value, far below 2^{63}.

If a check fails, the program falls back to a greedy proposal of B, also computed in floating point and checked in the same way, and it stops if that fails too. The result does not depend on which proposal is used: the children of a graph do not depend on the certificate (Lemma 3.5), and neither do $\operatorname{rk} M = |B|$ and $\det \Lambda(X)$.

Admissibility is a set of integer identities and inequalities, tested for all $2^r - 1$ nonzero $\mathbf{s}$ at once: the program computes $\mathbf{s}^T C$ and $\mathbf{s}^T \mathrm{adj}(G)\,\mathbf{s}$ for all these $\mathbf{s}$ in 64-bit integers. These are small as well: the entries of C are at most 2^{11} in absolute value, since $r \le 8$, those of $\mathrm{adj}(G)$ are at most 2^7 and those of M at most 2, so the integers of the test are at most 2^{14}, and $D \le 2^r \le 2^8$ by

Hadamard's inequality. When $r \geq 6$, the program computes $\det \Lambda$ by Lemma 2.4, as a subgroup closure in $(\mathbb{Z}/D\mathbb{Z})^r$, in the integers of Python, and checks that it is one of the three values $r+1$, 4, $9-r$ allowed by Lemma 2.2; when $r \leq 5$ the graph is not exceptional, by Proposition 2.6, and $\det \Lambda$ is not computed. The program also records, for every graph of the levels, the number of its nonzero admissible vectors, which gives the maximal exceptional graphs (Section 4).

Canonical forms

Canonical forms are those of `labelg` of nauty 2.9.3 [16], in its default mode (dense nauty, no `-S` or `-t`), and the graph6 strings (Appendix C) are sorted in byte order (`LC_ALL=C sort`). The canonical form depends on the program and its version, so a user with another canonical labelling should relabel the files before comparing strings. Duplicates are removed with `sort -u` on canonical strings, which is correct by the definition of a canonical form. Each of the worker processes of `grow.py`, which share the graphs of a level, pipes the children of its graphs through `labelg -q | sort -u` under `bash -o pipefail` and checks the exit status of the pipeline, so that a failure of `labelg` stops the program.

Software

We used Python 3.12 [19] with NumPy 2.5 [13] and SciPy 1.18 [21], and nauty 2.9.3 [16]; for the checks, PARI 2.17.3 [18] through cypari2 2.2.4 [9], and SageMath 10.9 [20].

B The checks of the program and of its output

Since the computation is large, we made a few sanity checks of the program and of its output; the scripts are in the repository, with the database. For $n \leq 10$, an exact test of every connected graph on n vertices from nauty's `geng -c` finds exactly the graphs of $\mathcal{L}_n$. A second program rereads the files and checks, with PARI [18], that every listed graph is exceptional, in the right list and listed once. A SageMath script finds the seven exceptional strongly regular graphs [2, Theorem 3.12.4] in list (A), and none of 34 small generalized line graphs in the lists.

Comparison with the literature

List (B) has, on 6, 7 and 8 vertices, the numbers of graphs that [8, §2.3 and Table A2] and [12] give for the exceptional graphs with $\lambda_{\min} > -2$ (Section 1), and the numbers of maximal exceptional graphs (Section 4) agree with [7] and [6]. Cvetković continues the passage of [6] quoted in Section 1 with "Minimal exceptional graphs, all having 6 vertices, were found some 25 years ago" (see also [8]). Every exceptional graph on 6 vertices is minimal, since an exceptional graph has at least 6 vertices by Proposition 2.6; thus the minimal exceptional graphs are exactly the exceptional graphs on 6 vertices, and there are 20 of them in list (B), as [6] states.

Cvetković also writes that "Most of the maximal exceptional graphs (432 of them) are 29 vertex cones" [6]. The maximal exceptional graphs are divided in [8, Chapter 6], as reported in [10, §3], into three types: (a) those with largest valency 28, the largest valency possible [4, Theorem 4.3], and exactly 29 vertices; (b) those with largest valency 28 and more than 29 vertices; and (c) those with largest valency less than 28. De Bruyn and van Veluw state that there are 430 graphs of type (a), 37 of type (b) and 6 of type (c); a graph of type (a) is a cone (a graph with a vertex adjacent to

all the others), since its vertex of valency 28 is adjacent to all the others. Our counts agree exactly: 430 of our maximal exceptional graphs have 29 vertices and a vertex of valency 28, 37 have more than 29 vertices and a vertex of valency 28, and 6 have largest valency less than 28. Two of the last 6 have 29 vertices, and both have largest valency 26:

```
\??GWWJ@}[ThijhfxLuIzom}‘vx{]AyXdYijTeUxrLNEiizKkrnS[N^}@~~{~~~~Z~~~{
\@G?GGBEDkLp\R[jthtTZkY|hfyk}C[xeTYjcmUyYlNHUizS[rna[N^~B~~}N^~~N~~~{
```

Thus exactly 430 of the 432 maximal exceptional graphs on 29 vertices are cones: the number 432 of [6] is the number of all maximal exceptional graphs on 29 vertices.

The program `count_regular.py` finds 187 regular graphs in the lists, all in list (A), in agreement with the 187 regular exceptional graphs of Bussemaker, Cvetković and Seidel [3] (see [2, §3.11] and [10]). With k the valency, 163 of them have $n = 2(k+2)$, 21 have $n = 3(k+2)/2$ and 3 have $n = 4(k+2)/3$, as they must by [2, Theorem 3.12.2]. Hoffman's ratio bound on the independence number of a k-regular graph on n vertices with least eigenvalue -2 is $2n/(k+2)$, so these are the classes of 163, 21 and 3 regular exceptional graphs of [10], with ratio bounds 4, 3 and $8/3$.

C The database

The files are in the directory `exceptional/database/` of the repository: `minus2/nNN.g6` holds list (A), `above-minus2/nNN.g6` list (B) and `maximal/nNN.g6` the maximal exceptional graphs, on `NN` vertices, where `NN` is n written with two digits. Each file has one graph6 string per line, sorted in byte order (Appendix A), and it is compressed with gzip (with suffix `.gz`) when larger than 20 MB; the three directories take about 770 MB in all. A graph6 string is the compact text format for graphs of nauty [16]; SageMath [20] reads one with `Graph(s)`, and the tools of nauty read the files directly, but not the gzipped ones, which have to be decompressed first, as in `zcat minus2/n22.g6.gz | countg`. The file `counts.json` records, for each n, the numbers of graphs in the three lists, the size of $\mathcal{S}_n$ and the lattice types of its graphs, the names of the files, and the time of the growth step and that it is complete; Table 1 is made from it. The `README.md` there repeats the definitions and gives the commands.

What the files contain

Each list stores its graphs on n vertices in one file; the file of list (A) on 19, 20 or 21 vertices is stored in two parts (below), which we will take together; and an absent file means that the list has no graphs on n vertices.

Theorem C.1. *Let $n \geq 1$. The strings in the file of list (A) on n vertices are the graph6 strings of the canonical forms of the exceptional graphs on n vertices with $\lambda_{\min} = -2$, one for each isomorphism class, and the strings in the file of list (B) on n vertices are those of the exceptional graphs on n vertices with $\lambda_{\min} > -2$, one for each isomorphism class. List (B) is empty unless $n \in \{6, 7, 8\}$, and both lists are empty for $n \leq 5$ and $n \geq 37$.*

Proof. As in the proof of Theorem 1.1, the exceptional graphs on n vertices are the graphs in $\mathcal{S}_n$ with $(\operatorname{rk} M, \det \Lambda) \in \{(6,3),(7,2),(8,1)\}$, and $\mathcal{L}_n$ consists of exactly one graph from each class of $\mathcal{S}_n$; the program computes, for each $X \in \mathcal{L}_n$, $\operatorname{rk} M(X)$ and, when it is at least 6, $\det \Lambda(X)$. It writes the graph6 string of X, which is a canonical form, to the file of list (A) if X is exceptional

with $\operatorname{rk} M < n$, to the file of list (B) if X is exceptional with $\operatorname{rk} M = n$, that is, with M positive definite, and to neither file otherwise. The bounds on n are those of Theorem 1.1. □

The list of maximal graphs holds the exceptional graphs for which the recorded number of nonzero admissible vectors is 0. By Theorem C.1 and Proposition 3.7, its strings are the graph6 strings of the canonical forms of the maximal exceptional graphs, one for each isomorphism class.

The two parts of the files on 19, 20 and 21 vertices

So that every file stays under 100 MB, each of the files of list (A) on 19, 20 and 21 vertices is split in two by number of edges, at the lower median m_0 of the numbers of edges: one part holds the graphs with at most m_0 edges and the other those with at least $m_0 + 1$ edges, each in byte order, and the file names give the bounds:

n	m_0	part	graphs
19	92	`minus2/n19.edges_le_92.g6.gz`	4 348 448
19	92	`minus2/n19.edges_ge_93.g6.gz`	3 939 179
20	103	`minus2/n20.edges_le_103.g6.gz`	4 047 472
20	103	`minus2/n20.edges_ge_104.g6.gz`	3 676 989
21	115	`minus2/n21.edges_le_115.g6.gz`	3 392 738
21	115	`minus2/n21.edges_ge_116.g6.gz`	2 834 489

The script `split_by_edges.py` writes the parts, with a fixed gzip header so that a rerun gives identical bytes, and checks that they merge back to the whole list line for line. The checks of Appendix B read the two parts together, through the function `read_list` of `e8common.py`, and `verify.py` also checks that every graph in a part has its number of edges in the range of the part.

Regenerating and verifying

We will give the commands as they are run in the directory `exceptional/database`; the number 13 in them is the number of worker processes. To regenerate everything, run

```
scripts/run_levels.sh WORKDIR DBDIR 1 37 13
```

with nauty and Python (NumPy, SciPy) on the path. Here `WORKDIR` is a scratch directory, `DBDIR` is the directory in which the script writes the lists and `counts.json`, and `1 37` asks for the growth steps from n to $n+1$ vertices for $1 \le n < 37$. With `.` as `DBDIR` the script overwrites the stored files; to compare a rerun with them, give an empty directory instead. The script writes the levels $\mathcal{S}_n$ to `WORKDIR/levels/nNN.g6` and their classification (rank, lattice determinant and number of nonzero admissible vectors of each graph) to `WORKDIR/levels/infoNN.txt`; a full run takes about 45 minutes on 13 cores, and the levels take 1.7 GB, or 3.8 GB with their classification. Then run `python scripts/split_by_edges.py DBDIR/minus2/nNN.g6.gz DBDIR/minus2` for $n = 19$, 20 and 21, and delete the three whole files. To verify the lists and to repeat the checks of Appendix B, run

```
python scripts/verify.py . 13
python scripts/check_bruteforce.py n WORKDIR/levels/nNN.g6 13
sage scripts/check_named.sage . WORKDIR/levels
python scripts/count_regular.py . 13 OUTFILE
```

where `verify.py` also needs cypari2, and `check_bruteforce.py` is meant for small n.

D The formalization of [4] in Lean

We have formalized [4] in the proof assistant Lean 4 [11], on top of its mathematical library Mathlib [15], in the directory `LineGraphs/` of the repository. A result is in Lean when its statement and its proof are written in Lean and the proof is accepted by the kernel of Lean, a small program that accepts a step only if it follows from the axioms of the logic and from what was proved before. Lean checks the proofs; whether the statements say what the paper says is left to the author. Every numbered result of [4] is in Lean, apart from Remark 2.11, which only points to a characterization proved in another paper, and the library contains no unproved step and no added axiom.

Every definition, lemma, proposition and theorem of Sections 2 and 3 is in Lean (Table 2), and so are the results of [4] that they use: [4, Theorem 3.5] (`LineSystem.classification`), [4, Theorem 4.2] (`SimpleGraph.isRepresentedIn_Dn_iff`), [4, Theorem 4.3] (`SimpleGraph.IsRepresentedIn.card_le_36`) and [4, Theorem 4.10] (`SimpleGraph.hoffman`). The library gives $\lambda_{\min}(X) \geq -2$ as the positive semidefiniteness of $M(X)$ over the reals, $\lambda_{\min}(X) > -2$ as positive definiteness, and $\det \Lambda(X)$ as the Gram determinant of a $\mathbb{Z}$-basis of the $\mathbb{Z}$-span of a representation, which it proves to depend neither on the basis nor on the representation. Of Theorems 1.1 and C.1, the library proves the mathematical content: that the exceptional graphs on n vertices are the graphs of the level $\mathcal{S}_n$ whose pair $(\operatorname{rk} M, \det \Lambda)$ is $(6,3)$, $(7,2)$ or $(8,1)$; that those with $\lambda_{\min} = -2$ are those with $\operatorname{rk} M < n$; that the growth with this classification selects one graph from each isomorphism class of each of the three lists; and the bounds on n. The mathematical content of the counts of Section 4 is Theorem 3.6 and Proposition 3.7. Not in Lean are the statement of Theorem C.1 about the files, the correctness of the program of Appendix A and of the canonical form of `labelg`, and the content of Appendices A to C: checking the 54 732 619 strings in the kernel is out of reach.

result	Lean declaration
exceptional graph (Section 1)	`SimpleGraph.IsExceptional`
$\det \Lambda(X)$ (Section 2)	`SimpleGraph.latticeDet`, `SimpleGraph.IsRepresentation.latticeDet_eq_det_gram`
Lemma 2.1	`SimpleGraph.IsRepresentation.latticeRoots`, `SimpleGraph.IsRepresentation.isStarClosed_latticeRoots`, `SimpleGraph.IsRepresentation.isIndecomposable_latticeRoots`, `SimpleGraph.IsRepresentation.span_latticeRoots`, `SimpleGraph.IsRepresentation.isRepresentedIn_latticeRoots`
Lemma 2.2	`SimpleGraph.IsRepresentation.exists_unique_root_lattice`
Lemma 2.3	`SimpleGraph.latticeDet_eq_prod_smithNormalForm`
Lemma 2.4	`SimpleGraph.latticeDet_eq_div_card_basisDetGroup_sq`
Lemma 2.5	`LineSystem.E`$_6$`.not_exists_lattice_isometry_int`, `LineSystem.E`$_6$`.not_exists_integral_gram_simpleRoot`
Proposition 2.6 (1)	`SimpleGraph.IsRepresentation.isGeneralizedLineGraph_iff_rootType_isClassical`
Proposition 2.6 (2)	`SimpleGraph.isExceptional_iff_rank_latticeDet`
Proposition 2.6 (3)	`SimpleGraph.isRepresentedIn_E`$_8$`_iff_posSemidef_and_rank_le`
Lemma 3.1	`SimpleGraph.Connected.exists_noncut_isRepresentedIn_E`$_8$
Definition 3.2	`Matrix.IsGramCertificate`
Lemma 3.3	`SimpleGraph.exists_integralGramMatrix_certificate_iff`, `Matrix.IsGramCertificate.card_eq_rank`, `Matrix.isGramCertificate_of_linearIndependent_rows`
Lemma 3.4	`SimpleGraph.isRepresentedIn_adjoinVertex_iff`
Lemma 3.5	`SimpleGraph.isRepresentedIn_adjoinVertex_iff_exists_admissible`, `SimpleGraph.existsUnique_admissible_of_isRepresentedIn_adjoinVertex`
Theorem 3.6	`SimpleGraph.growth_represents_levels`, for every canonical form and every choice of certificates; `SimpleGraph.CanonicalForm.ofChoice` shows that a canonical form exists
Theorems 1.1 and C.1, the classification	`SimpleGraph.isExceptional_and_not_posDef_iff`, `SimpleGraph.isExceptional_and_posDef_iff`, `SimpleGraph.IsExceptional.card_bounds`, `SimpleGraph.IsExceptional.card_mem_of_posDef`, `SimpleGraph.singularExceptionalGrowth_represents`, `SimpleGraph.definiteExceptionalGrowth_represents`
Proposition 3.7, and the list of maximal graphs	`SimpleGraph.IsExceptional.isMaximalExceptional_iff_no_admissible`, `SimpleGraph.IsExceptional.isMaximalExceptional_iff_no_extension`, `SimpleGraph.maximalExceptionalGrowth_represents`

Table 2: The Lean statements of the results of this note, in the directory `LineGraphs/Exceptional/`, apart from those of Lemma 2.1, which are in `LineGraphs/Seidel/LatticeRoots.lean`; in the names, the subscripts of E_6 and E_8 are Unicode subscript digits.

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